Topic 14: Switch Node & Workflow Routing — Requirements Document

1. Introduction

This document defines the functional and technical requirements for implementing multi-branch workflow routing using the n8n Switch node.

2. Objectives and Scope

2.1 Course Objective

Teach trainees how to configure the Switch node to route items through multiple output branches based on different conditions.

2.2 In Scope

- Adding and configuring the Switch node.
- Using dynamic expressions such as `{{ $json.role }}`.
- Configuring multiple string comparison rules.
- Configuring a default fallback output.
- Connecting downstream Edit Fields nodes.
- Testing multiple input items.

2.3 Out of Scope

- Custom JavaScript switch statements.
- Dynamic generation of output branches at runtime.

3. Target Audience and Prerequisites

- Trainees and employees.
- Completion of Topics 1–13.
- Understanding of basic comparison operators.
- Basic knowledge of n8n IF node branching.

4. Functional Requirements

Requirement ID	Description	Priority
REQ-001	Include a Switch node to route records based on property matching.	High
REQ-002	Configure admin, editor, and viewer comparison rules.	High
REQ-003	Configure a default fallback output.	High
REQ-004	Evaluate rules in sequential top-to-bottom order.	High
REQ-005	Read the role dynamically using `{{ $json.role }}`.	High
REQ-006	Connect four downstream Edit Fields nodes for Admin, Editor, Viewer, and Guest.	High
REQ-007	Process multiple profiles simultaneously and route them correctly.	High

5. Branch Mapping

Output	Destination Node	Field
Output 0	Set Admin Status	isAdmin = true
Output 1	Set Editor Status	isEditor = true
Output 2	Set Viewer Status	isViewer = true
Fallback	Set Guest Status	isGuest = true

6. Technical Requirements

ID	Requirement	Description
TR-001	Index-Based Branch Routing	Each matching rule routes the item through its corresponding output connector.
TR-002	Dynamic Array Partitioning	Multiple JSON profiles are split and routed into separate output groups.

7. Stakeholders

Persona	Role	Responsibility
Trainee / Employee	Student	Implement and test the workflow.
Instructor / Reviewer	Evaluator	Verify routing rules, fallback configuration, and outputs.

8. Assumptions and Constraints

- Input data contains a role property.
- Role values match predefined rules or use fallback.
- The output configuration is static.
- Adding or changing rules may require reconnecting output paths.

9. Acceptance Criteria

The workflow is accepted when:

- Three comparison rules are configured.
- The fallback output is enabled.
- Four downstream nodes are connected.
- Four sample profiles are routed correctly.
- Each downstream node returns the expected Boolean value.